Research in Mathematics: Real Analysis

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Declaration

I hereby declare that this project report is my own work and has not been submitted to any other university or institution for any degree or diploma. All sources used are acknowledged in the bibliography.

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Chapter 1

Number Systems

1.1 Introduction

This chapter introduces the real number system. This topic is discussed in Rudin [1] and forms a basic part of real analysis.

1.2 Main Content

We study ordered sets and the least upper bound property. This topic is discussed in Rudin [1] and forms a basic part of real analysis.

1.3 Examples

The rational numbers do not have this property. This topic is discussed in Rudin [1] and forms a basic part of real analysis.

Chapter 2

Basic Topology

2.1 Introduction

This chapter introduces metric spaces. This topic is discussed in Rudin [1] and forms a basic part of real analysis.

2.2 Main Content

We study open sets and compact sets. This topic is discussed in Rudin [1] and forms a basic part of real analysis.

2.3 Examples

Closed and bounded intervals are compact subsets of the real line. This topic is discussed in Rudin [1] and forms a basic part of real analysis.

Chapter 3

Sequences and Series

3.1 Introduction

This chapter introduces numerical sequences. This topic is discussed in Rudin [1] and forms a basic part of real analysis.

3.2 Main Content

We study convergence and Cauchy sequences. This topic is discussed in Rudin [1] and forms a basic part of real analysis.

3.3 Examples

The sequence $1/n$ converges to zero. This topic is discussed in Rudin [1] and forms a basic part of real analysis.

Chapter 4

Continuity

4.1 Introduction

This chapter introduces limits of functions. This topic is discussed in Rudin [1] and forms a basic part of real analysis.

4.2 Main Content

We study continuous functions on metric spaces. This topic is discussed in Rudin [1] and forms a basic part of real analysis.

4.3 Examples

Polynomial functions are continuous on the whole real line. This topic is discussed in Rudin [1] and forms a basic part of real analysis.

Chapter 5

Conclusion

In this report we studied the basic ideas of real analysis, namely number systems, topology, sequences and continuity, following Rudin [1]. These ideas form the foundation for further study of differentiation and integration.

Bibliography

[1] W. Rudin, *Principles of Mathematical Analysis*, 3rd ed., McGraw-Hill, 1976.